

SE LAB 01 Submission pdf

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1. Requirements Table

The requirements table below contains exactly five Functional Requirements (FR-001 to FR-005) and two Non-Functional Requirements (NFR-001 & NFR-002), including pre-defined requirements from the prompt.

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale
FR-001	Functional	The system shall lock buyer payment in escrow and transition order state to 'Under Inspection' until an inspector submits physical verification sign-off.	High	Pass: Funds held securely and released only on dual verification; Fail: Funds released prematurely to seller.	Core escrow and inspection workflow security.
FR-002	Functional	The system shall allow an inspection agent to upload physical inspection	High	Pass: Report is successfully uploaded, stored, and viewable by buyer/seller; Fail: Report	Required for objective product condition verification.

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		reports and condition verification certificates.		upload fails or is inaccessible.	
FR-003	Functional	The system shall allow the buyer to submit a final delivery receipt sign-off once the inspected item is received to trigger fund release to the seller.	High	Pass: Funds transfer from escrow to seller account upon buyer sign-off; Fail: Funds remain locked or release without sign-off.	Ensures buyer satisfaction and finalizes transaction safely.
FR-004	Functional	The system shall allow a buyer or seller to raise a dispute ticket regarding product condition or delivery issues.	Medium	Pass: Ticket is generated with a unique ID and status set to 'Active Investigation'; Fail: Ticket creation fails.	Necessary for handling transaction conflicts and disputes.

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale
FR-005	Functional	The system shall notify all relevant stakeholders (Buyer, Seller, Inspection Agent) via email/SMS upon any status change in the order or escrow state.	Medium	Pass: Notifications are dispatched within 5 seconds of state change; Fail: Stakeholders are not notified.	Keeps participants informed of critical order milestones.
NFR-001	Non-functional (Performance & Security)	Dispute resolution tickets must freeze escrow funds immediately and prevent account withdrawals during active investigation .	High	Pass: Benchmarking tests confirm target latency and security standards under simulated peak load; Fail: Funds remain accessible during dispute.	Protects funds during financial disputes.

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NFR-002	Non-functional (Security & Reliability)	The platform must encrypt all financial transaction data and personal user information using AES-256 standards at rest and TLS 1.3 in transit.	High	Pass: Security compliance scans verify AES-256 and TLS 1.3 encryption protocols across all endpoints; Fail: Unencrypted data transmission detected.	Ensures data privacy and regulatory compliance for financial transactions.

2. UML Use-Case Model Specification

To model this system in a tool like Draw.io or Lucidchart, use the following structure:

Actors

- **Buyer:** Initiates purchases, funds escrow, signs off on delivery, and can raise disputes.
- **Seller:** Lists products and receives funds upon successful verification and sign-off.
- **Inspection Agent:** Conducts physical verification and submits sign-offs.
- **Payment Gateway (External System):** Processes secure transactions and holds funds in escrow.

Use Cases

- **UC-01:** Browse Products & Place Order
- **UC-02:** Lock Payment in Escrow (Includes payment processing)
- **UC-03:** Conduct Physical Inspection & Verify Condition

- **UC-04:** Sign-Off and Release Funds
- **UC-05:** Raise Dispute Ticket (Extends UC-04 or UC-03)

Relationships

- **<<include>>:** UC-01 (Place Order) includes Process Payment.
- **<<extend>>:** UC-05 (Raise Dispute Ticket) extends UC-04 (Sign-Off and Release Funds) when a buyer rejects the item condition or delivery.

3. Use-Case Flow Specification Document

Use Case: UC-04: Sign-Off and Release Funds

- **Preconditions:**
 1. Buyer has locked payment in escrow.
 2. The inspection agent has successfully submitted physical verification sign-off.
 3. The order state is marked as 'Inspected and Ready for Delivery'.
- **Postconditions:**
 1. Escrow funds are released to the seller's verified account.
 2. Order status updates to 'Completed'.

Main Success Scenario

1. Buyer receives the physical product shipped from the inspection facility.
2. Buyer logs into the marketplace and navigates to active orders.
3. Buyer selects the option to review product condition and confirm receipt.
4. System prompts the buyer to provide final delivery sign-off.
5. Buyer confirms and signs off on the product condition.
6. System verifies dual verification (Inspector sign-off + Buyer sign-off).
7. System triggers the payment gateway to release funds from escrow to the seller.
8. System updates the order state to 'Completed' and issues digital receipts to both parties.
9. Use case ends successfully.

Alternate Flow

- **6a. Buyer Rejects Item Condition / Delivery Issue**
 - **6a1.** Buyer clicks "Raise Dispute" instead of signing off.
 - **6a2.** System triggers **NFR-001**, immediately freezing escrow funds and locking account withdrawals for the active investigation.
 - **6a3.** System creates a dispute ticket, assigns it to support staff, and notifies all stakeholders.
 - **6a4.** Use case branches into the Dispute Resolution workflow.